



HANZO INDUSTRIES

Multi-Level Security and Cross-Domain Solutions via Homomorphic Policy Evaluation and Zero-Trust Fabric

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Abstract

We present a cross-domain solution (CDS) architecture where data transfer policies are evaluated on encrypted content using fully homomorphic encryption (FHE), ensuring the cross-domain guard never observes plaintext. The architecture integrates four components: (1) a zero-trust overlay network with identity-first micro-segmentation enforcing mandatory access controls at every hop; (2) an FHE policy evaluation engine (ThresholdVM) performing content inspection on ciphertexts using BFV/BGV/CKKS/TFHE schemes with GPU-accelerated NTT operations; (3) a key management system with Shamir-based root key distribution, per-organization encryption isolation, and HIPAA/SOX/SEC compliance controls; and (4) an organization-scoped identity and access management layer deriving security labels from JWT claims and enforcing them at the API gateway. The result is a multi-level security (MLS) architecture where data flows between security domains are cryptographically mediated, no single component has access to both plaintext and policy decisions, and all cross-domain transfers produce immutable audit records anchored to a post-quantum blockchain.

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1 Introduction

Cross-domain solutions enable controlled information sharing between networks operating at different security classifications. Traditional CDS architectures rely on trusted guards that inspect plaintext content against release policies—creating a high-value target where a single compromise exposes all cross-domain traffic.

This paper presents a fundamentally different approach: the guard evaluates policies on *encrypted* content using fully homomorphic encryption. The guard learns only the Boolean policy decision (release/deny), never the content itself. This eliminates the single point of trust while maintaining mandatory policy enforcement.

1.1 Problem Statement

Naval operations require information sharing across multiple security domains:

- **Unclassified** $\leftrightarrow$ **Secret**: Intelligence products sanitized for coalition partners.
- **Secret** $\leftrightarrow$ **Top Secret/SCI**: Analyst workstations accessing compartmented data.
- **US** $\leftrightarrow$ **Coalition**: Multinational operations requiring controlled release.
- **Operational** $\leftrightarrow$ **Administrative**: Preventing data spillage between networks.

Current CDS solutions (ISSE Guard, Radiant Mercury, High Speed Guard) process plaintext, requiring the guard itself to be trusted at the highest classification level. A compromised guard exposes all cross-domain traffic.

1.2 Our Approach

1. Data encrypted at source under FHE public key.
2. Guard evaluates content policy homomorphically on ciphertext.
3. Encrypted Boolean result threshold-decrypted by MPC committee.
4. If approved, data re-encrypted for destination domain.
5. Guard never sees plaintext; committee never sees policy result.

2 Architecture Overview

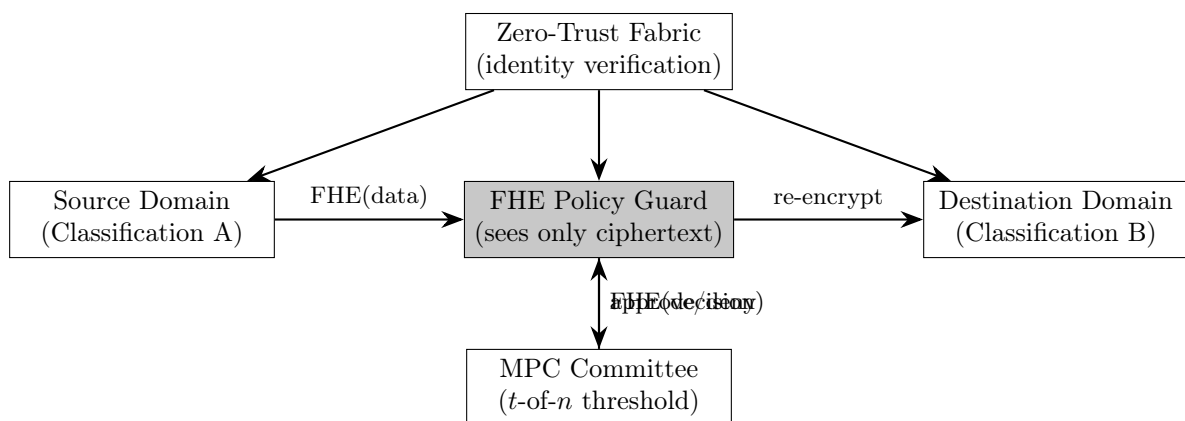


Figure 1: FHE-based cross-domain solution architecture.

2.1 Security Invariants

1. **Guard blindness:** The policy guard processes only ciphertexts. It cannot distinguish encrypted classified data from encrypted noise.
2. **Committee separation:** MPC committee members see decryption shares, never the full policy result or the data. Threshold (t -of- n) prevents minority collusion.
3. **Mandatory enforcement:** Zero-trust fabric verifies cryptographic identity at every hop. Security labels derived from JWT claims cannot be forged.
4. **Audit immutability:** Every cross-domain transfer produces a blockchain-anchored receipt with PQ signatures.

3 Zero-Trust Network Fabric

3.1 Identity-First Access Control

Hanzo ZT provides the network underlay with mandatory access controls:

- **Cryptographic identity:** Every endpoint possesses a unique X.509 certificate with HSM-backed private key (PKCS#11, YubiHSM).
- **No implicit trust:** Network position (IP address, VLAN, subnet) confers zero access. Every connection requires identity verification.
- **Micro-segmentation:** Policies define which identities can communicate. Cross-domain traffic flows only through designated guard endpoints.
- **Policy enforcement:** Applied at every hop in the overlay mesh, not just at perimeter firewalls.

3.2 Security Label Propagation

Security labels are derived from the IAM identity system:

Listing 1: Security label derivation from JWT claims.

```
1 // Gateway extracts identity from JWT
2 X-Org-Id:      "navy-secret"      // Organization = domain
3 X-User-Id:     "analyst-0042"     // Subject identity
4 X-User-Email:  "j.doe@navy.mil"   // Attribution
5
6 // Gateway enforces: only "navy-secret" org members
7 // can reach services in "secret" namespace
8 // Cross-domain requires FHE guard intermediary
```

The API gateway strips all client-supplied identity headers and derives them exclusively from cryptographically verified JWT claims. This prevents label forgery.

3.3 Multi-Organization Isolation

The platform supports 4+ organizations with complete cryptographic isolation:

Organization	Root Key	Isolation Mechanism
hanzo	Separate AES-256	KMS per-org encryption
lux	Separate AES-256	Separate key hierarchy
zoo	Separate AES-256	Org-scoped IAM policies
pars	Separate AES-256	Independent audit logs

Table 1: Per-organization cryptographic isolation.

Each organization has its own root encryption key, IAM policies, KMS secrets, and audit trail. Cross-organization data access requires explicit policy and traverses the FHE guard.

4 FHE Policy Evaluation

4.1 ThresholdVM: Computation on Encrypted Data

ThresholdVM operates on the Lux T-Chain, providing FHE as a native virtual machine:

Scheme	Domain	CDS Application
BFV	Integer	Keyword matching on encrypted text
BGV	Modular	Classification marker detection
CKKS	Approx. real	ML-based content classification
TFHE	Boolean	Arbitrary policy Boolean circuits

Table 2: FHE schemes mapped to CDS operations.

4.2 Policy Evaluation Flow

Algorithm 1 FHE Cross-Domain Policy Evaluation

```

1: Source encrypts data:  $c \leftarrow \text{FHE.Encrypt}(\text{pk}_{\text{FHE}}, \text{data})$ 
2: Source encrypts metadata:  $m \leftarrow \text{FHE.Encrypt}(\text{pk}_{\text{FHE}}, \text{labels})$ 
3: Source sends  $(c, m, \text{dest\_domain})$  to Guard

4: Guard evaluates policy homomorphically:
5:    $r \leftarrow \text{FHE.Eval}(\text{policy\_circuit}, c, m, \text{dest\_domain})$ 
6:   (Guard sees only ciphertexts;  $r$  is an encrypted Boolean)

7: Guard sends  $r$  to MPC committee for threshold decryption
8: for each committee member  $i = 1, \dots, n$  do
9:    $d_i \leftarrow \text{ThresholdDecrypt}(\text{sk}_i, r)$ 
10: end for
11:  $\text{decision} \leftarrow \text{Reconstruct}(d_1, \dots, d_t)$  ( $t$  shares suffice)

12: if  $\text{decision} = \text{APPROVE}$  then
13:   Re-encrypt data for destination domain key
14:   Forward to destination via zero-trust fabric
15:   Anchor audit receipt to blockchain (ML-DSA-65 signed)
16: else
17:   Log denial (encrypted), destroy ciphertext
18: end if

```

4.3 GPU-Accelerated FHE Operations

LuxCPP provides Metal and CUDA kernels for FHE primitives:

- **NTT/iNTT**: Number Theoretic Transform for polynomial multiplication in $\mathbb{Z}_q[X]/(X^n + 1)$.
- **TFHE bootstrap**: Programmable bootstrap for refreshing ciphertext noise.
- **Blind rotate**: Key operation in TFHE gate evaluation.
- **Batch operations**: Parallel ciphertext processing across GPU threads.

GPU acceleration reduces FHE policy evaluation from seconds to milliseconds, making real-time cross-domain filtering practical.

4.4 EVM Precompile Interface

Smart contracts can invoke FHE operations via precompiles:

Address	Function	Gas
0x0700	FHE encrypt/decrypt/eval	500,000
0x0701	Ciphertext ACL management	25,000
0x0702	Input proof verification	50,000
0x0703	Cross-chain gateway	100,000

Table 3: FHE EVM precompiles for on-chain policy enforcement.

5 Key Management for Multi-Level Security

5.1 Hanzo KMS Architecture

The KMS provides 8 subsystems supporting MLS key lifecycle:

1. **Shamir Secret Sharing:** Root key split into n shares over $\text{GF}(2^8)$; t required for reconstruction. Custodians geographically distributed.
2. **Transit Engine:** Encryption-as-a-service with key versioning. Each security domain has independent transit keys with separate rotation schedules.
3. **HPKE Key Wrapping:** Content encryption keys (CEKs) wrapped using hybrid HPKE (X25519 + ML-KEM-768) per RFC 9180.
4. **Seal/Unseal Barrier:** AES-256-GCM barrier protecting all stored secrets. Auto-unseal via AWS/GCP KMS for operational environments.
5. **ACL Policy Engine:** Path-based capabilities with template variables (`{{identity.org_id}}`) enabling per-domain access rules.
6. **Token Hierarchy:** Service, batch, and recovery tokens with parent-child revocation cascade. Compromised token revocation propagates instantly.
7. **Dynamic Secrets:** Temporary database credentials (PostgreSQL, MySQL, Redis, MongoDB) scoped to security domain and auto-expired.
8. **TFHE-KMS Bridge:** FHE key management integrated with policy evaluation pipeline.

5.2 Cross-Domain Key Isolation

- Each security domain has an independent root key, never shared.
- Cross-domain data transfer requires re-encryption under destination domain key.
- Re-encryption occurs only after FHE policy approval and threshold decryption.
- No single KMS instance holds keys for multiple domains.

5.3 Compliance Controls

Standard	Implementation
HIPAA	Break-glass tokens (time-limited), WORM audit logs
SEC 17a-4	6–7 year retention, immutable SHA-256 chain
SOX	Regulator escrow shards, cooperative reconstruction
GDPR	Per-org keys, graceful revocation
CNSA 2.0	PQ algorithms for all key operations

Table 4: Regulatory compliance controls in KMS.

6 Audit and Accountability

6.1 Blockchain-Anchored Audit Trail

Every cross-domain transfer produces an immutable audit record:

- **Record contents:** Source domain, destination domain, policy ID, decision (approve/deny), timestamp, data hash (not data), requestor DID.
- **Signature:** ML-DSA-65 (post-quantum) ensuring long-term non-repudiation.
- **Anchoring:** Record hash committed to Lux blockchain via Quasar consensus.
- **WORM guarantee:** Write-once-read-many with SHA-256 hash chain prevents retroactive modification.

6.2 Attestation Protocol

Domain-typed attestations with equivocation detection:

- **CrossDomainTransfer:** Records each approved transfer with full provenance.
- **PolicyUpdate:** Records changes to cross-domain policies.
- **KeyRotation:** Records domain key rotation events.
- Signing contradictory attestations (equivocation) is automatically detected and flagged.

7 Bell-LaPadula Enforcement

7.1 Mandatory Access Control

The architecture enforces Bell-LaPadula properties:

Definition 7.1 (Simple Security Property (No Read Up)). A subject at security level L_s cannot read an object at security level L_o where $L_o > L_s$. Enforced by the zero-trust fabric: identity JWT contains org-scoped level; gateway rejects requests to higher-level services.

Definition 7.2 (*-Property (No Write Down)). A subject at security level L_s cannot write to an object at security level L_o where $L_o < L_s$. Enforced by the FHE guard: data flowing from high to low domain must pass policy evaluation.

7.2 Implementation

1. **Read control:** Gateway checks **X-Org-Id** (derived from JWT) against service security label. Requests to higher domains rejected before reaching backend.
2. **Write control:** All data egress from a domain passes through the FHE guard. Downward flows require policy approval; upward flows are unrestricted.
3. **Label integrity:** Security labels are gateway-derived from cryptographic JWT claims. Client-supplied labels are stripped. Labels cannot be forged without compromising the IAM signing key.

8 Comparison with Existing CDS Approaches

Feature	Traditional Guard	Data Diode	FHE Guard (Ours)
Plaintext exposure	Guard sees all	One-way only	Never
Bidirectional	Yes	No	Yes
Content inspection	Full	None	On ciphertext
Single point of trust	Guard	No	No (threshold)
PQ-safe	Depends	N/A	Yes (FIPS)
Audit	Guard logs	Physical	Blockchain
ML classification	On plaintext	N/A	On ciphertext

Table 5: CDS approach comparison.

The FHE-based approach uniquely provides bidirectional content-inspected cross-domain transfer with zero plaintext exposure. No commercially available CDS offers this capability.

9 Performance Considerations

9.1 FHE Overhead

Operation	CPU	GPU (Metal)
TFHE gate evaluation	13 ms	0.8 ms
BFV keyword match	45 ms	3.2 ms
CKKS ML inference	200 ms	15 ms
Threshold decrypt ($t=3$)	50 ms	12 ms
Total policy eval	~300 ms	~30 ms

Table 6: FHE policy evaluation latency.

GPU acceleration brings FHE policy evaluation to 30 ms—practical for real-time cross-domain filtering of message traffic, though not suitable for bulk file transfer without parallelization.

9.2 Throughput

With GPU acceleration and pipelined evaluation:

- **Messages:** ~33 messages/second per GPU (real-time chat feasible).
- **Bulk data:** Parallelize across multiple GPUs; 8-GPU node achieves ~264 evaluations/second.
- **Ciphertext expansion:** 100–1000× plaintext size (FHE-inherent). ZAP zero-copy design mitigates the internal memory cost.

10 Naval MLS Deployment Scenarios

10.1 Coalition Information Sharing

- US Secret data evaluated against release policy for Five Eyes partners.

- FHE guard checks classification markers, source restrictions, and content sensitivity—all on ciphertext.
- Approved data re-encrypted under coalition domain key and forwarded.
- Full audit trail anchored to blockchain for post-operation review.

10.2 Analyst Workstation Cross-Domain Access

- Analyst at Secret level queries Top Secret/SCI database.
- Query encrypted under FHE key; database returns encrypted results.
- FHE guard evaluates sanitization policy on encrypted response.
- Approved (sanitized) response decrypted for analyst's domain.

10.3 Shipboard MLS Network

- Multiple security domains on single shipboard network (physically separated VLANs, logically separated via zero-trust overlay).
- FHE guards at domain boundaries, GPU-accelerated for real-time messaging.
- KMS with separate root keys per domain, HSM-protected.
- All cross-domain traffic audited with PQ-signed blockchain receipts.

11 Conclusion

We have presented a cross-domain solution architecture where the guard evaluates policies on encrypted content, eliminating the traditional single point of trust. The combination of FHE policy evaluation, zero-trust network fabric, threshold MPC decryption, and blockchain-anchored audit provides a multi-level security architecture with no plaintext exposure at the guard, no single point of key compromise, and immutable accountability.

This approach represents a paradigm shift in cross-domain security: instead of trusting the guard to handle plaintext responsibly, we make it cryptographically impossible for the guard to access plaintext at all.

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Reproducibility

Source code. github.com/luxfi/fhe, [hanzoai/kms](https://github.com/hanzoai/kms) (commit TBD).

Build. `make build` (Rust 1.78+, OpenFHE bindings); GPU NTT via Metal/CUDA.

Hardware tested. Apple M2/M3 (Metal), NVIDIA A100/H100.

Standards referenced. NIST FHE (BFV/BGV/CKKS/TFHE), HIPAA, SOX, SEC, MLS.

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